
RANDOM TRIANGLES ON CONCENTRIC CIRCLES

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ABSTRACT

Lewis Carroll’s Pillow Problem asks for the probability that three points chosen at random in the plane form an obtuse triangle. The question has no answer until the sampling scheme is pinned down, and the cleanest way to pin it down is to put the points on a circle, which gives $3/4$. This paper solves the version in which the three vertices lie on three concentric circles of radii r_1 , r_2 , and r_3 . The answer is a sum of three terms, each the probability that a weighted sum of two independent arcsine variables exceeds a threshold, and it collapses to $3/4$ when the radii are equal. Reading the formula gives closed forms in two subfamilies, one an arcsine and one the Legendre chi function; a Pythagorean condition deciding which vertex can carry the obtuse angle; and the bound $1/2 \leq P < 1$, with the minimum attained only when one vertex sits at the common center and the other two radii are equal. Conditioning on the radii shows that the same formula is the kernel for every independent rotationally symmetric sampling scheme, so the bound applies to all of them at once and averaging over Rayleigh radii recovers the known Gaussian value of $3/4$ in closed form, supplying a link between the circular and Gaussian conventions that the literature records as missing. Simulation confirms the results throughout.

Keywords geometric probability • concentric circles • Pillow Problem • random triangles • arcsine distribution

MSC 2020: 60D05 (primary); 52A22, 00A08 (secondary).

1 Introduction

Problem 58 of Carroll’s *Pillow Problems* (Carroll 1893) reads: “Three points are taken at random on an infinite plane. Find the chance of their being the vertices of an obtuse-angled triangle.” Carroll answered $3\pi/(8\pi - 6\sqrt{3}) \approx 0.639$, but the derivation conditions on the longest side in a way that quietly fixes a scale, and the problem as posed has no answer at all: there is no uniform distribution on the plane, and different limiting schemes give different numbers. Portnoy (1994) works through Carroll’s argument and the several defensible alternatives; Guy (1993) and Eisenberg and Sullivan (1996) reach the same conclusion from other directions. The situation is the Bertrand paradox in a different costume, and the standard resolution is the same one: state the sampling mechanism, then compute.

I met Buffon’s needle at the end of an undergraduate degree in statistics, through my advisor Harry Khamis, who had published two extensions of it: one replacing the ruled plane with a family of concentric circles (Khamis 1987), and one with radial lines (Khamis 1991). The first is where the present problem started. Buffon’s needle is the well-posed case: dropped on a ruled plane, its position and angle take their distribution from the symmetry of the strips (M. G. Kendall and Moran 1963, sec. 3.26). Carroll’s three points have no such symmetry to draw on, so the convention has to be supplied by hand. Concentric circles are what this paper supplies instead, and the choice is borrowed rather than invented.

One mechanism is unusually clean. Put the three points independently and uniformly on a common circle. A triangle inscribed in a circle is acute exactly when the center lies inside it, so the acute probability is the probability that three

uniform points on a circle contain the center, which is $1/4$ (Wendel 1962). The obtuse probability is therefore $3/4$, with no conditioning tricks and no limits.

A second route to the same number is more transparent still, and it will matter later. Discard the points and keep the angles. The triple (A, B, C) with $A + B + C = \pi$ ranges over an equilateral triangle in barycentric coordinates, and Edelman and Strang (2015) take the uniform distribution on it as the simplest model of a random triangle. The angle at a given vertex is obtuse exactly when it exceeds half the total, and that corner region is a copy of the whole simplex at half scale, hence a quarter of the area. The three corners are disjoint, so $P(\text{obtuse}) = 3/4$, and the acute triangles form the medial triangle, of area $1/4$ (Figure 1).

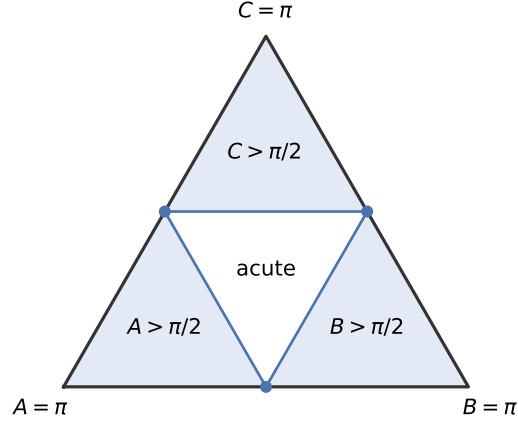


Figure 1: Angle space. Every triangle is a point of $\{A + B + C = \pi\}$, drawn in barycentric coordinates, and the uniform distribution on it is the simplest model of a random triangle. The angle at A exceeds $\pi/2$ in the shaded corner nearest the vertex $A = \pi$, and that corner is the whole picture at half scale, so it holds a quarter of the area. The same goes for B and C , and the three cannot overlap, leaving the medial triangle, again a quarter, as the acute triangles. Its edges are the right triangles and its corners are degenerate, with one angle equal to zero.

This is not a second convention but the first one in different coordinates. Rooted at one of the sampled points, the three arc gaps of three uniform points on a circle are uniform on $\{g_1 + g_2 + g_3 = 2\pi\}$, and the inscribed angle theorem makes the interior angle at each vertex half the arc opposite it. The angle triple is therefore uniform on $\{A + B + C = \pi\}$: up to labelling and reflection, the circular convention and the uniform-angle convention induce the same law on triangle shape. (The rooting matters. Gaps read off against a fixed external origin are size-biased and are not Dirichlet.)

With Buffon's needle still in mind, I came at the question a third way as a graduate student, and this is the version that convinced me. Buffon drops a needle of finite length ℓ ; drop needles of infinite length instead, which is to say lines, and three of them cut out a triangle (Figure 2). Toss them under the same invariant measure that makes Buffon's problem well posed, $dp d\theta$. Its angular part is uniform, so take the directions $\theta_1, \theta_2, \theta_3$ independent and uniform on $[0, \pi)$, and let the offsets be anything at all. The offset part is not normalizable, exactly as in Carroll's problem, but here that costs nothing: sliding a line parallel to itself moves and resizes the triangle without changing any of its angles, so the shape depends on the directions alone. Three lines fail to cut out a triangle only when two of them are parallel or all three pass through a common point, and both have probability zero. A right triangle needs an angle equal to $\pi/2$ exactly, which is also a null event, so obtuse and acute between them account for everything.

Sort the directions as $\theta_{(1)} \leq \theta_{(2)} \leq \theta_{(3)}$ in $[0, \pi)$. The interior angles are then

$$\theta_{(2)} - \theta_{(1)}, \quad \theta_{(3)} - \theta_{(2)}, \quad \pi - (\theta_{(3)} - \theta_{(1)}),$$

which are the three gaps between the directions on a circle of circumference π . A triangle is obtuse exactly when one angle exceeds $\pi/2$, that is when one gap covers more than half of that circle. Fix a direction and look at the gap running counterclockwise from it: that gap covers more than half the circle precisely when the other two directions both fall in the opposite half, which has probability $(\frac{1}{2})^2 = \frac{1}{4}$. Two gaps cannot both exceed half, so the three cases are disjoint and

$$P(\text{obtuse}) = 3 \times \frac{1}{4} = \frac{3}{4}.$$

Three constructions, then, and one measure. Uniform points on a circle, the uniform distribution on angle space, and lines under the invariant measure all induce the same law on the shape of a triangle.

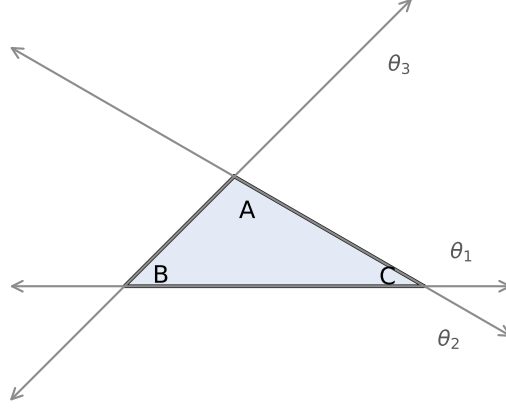


Figure 2: Three needles of infinite length, which is to say lines, tossed onto the plane and drawn with arrowheads to mark that they run on forever. They cut out a triangle. Its interior angles A, B, C are the gaps between the line directions $\theta_1, \theta_2, \theta_3$, so sliding any line parallel to itself moves and resizes the triangle but leaves all three angles alone.

A genuinely different convention reaches the same $3/4$ by an argument worth seeing on its own terms. Take the six vertex coordinates to be independent standard Gaussians. Shape theory then places the triangle uniformly on a hemisphere of shapes, a result of David Kendall and collaborators (D. G. Kendall 1989), who is not the M. G. Kendall of the paragraph above, and Edelman and Strang (2015) show (their Theorem 7) that under that law the squared side lengths, normalized so that $a^2 + b^2 + c^2 = 1$, are each uniform on $[0, 2/3]$. The angle opposite c is obtuse exactly when $c^2 > a^2 + b^2$, which under the normalization is simply $c^2 > 1/2$. So a given angle is obtuse with probability

$$\frac{2/3 - 1/2}{2/3} = \frac{1}{4},$$

and because at most one angle of a triangle can be obtuse, the three cases add to $3/4$.

Both moves in that argument reappear below. Additivity over the three vertices is Lemma 2.1. A Pythagorean threshold on squared quantities decides which vertex can carry the obtuse angle in Corollary 5.2, though there the inequality constrains the radii of the circles rather than the sides of the triangle.

The circle mechanism is the one this paper loosens, by giving each vertex its own circle: Khamis's substitution again, now applied to Carroll's three points rather than to Buffon's needle. Fix radii $r_1, r_2, r_3 \geq 0$ and let vertex k be uniform on the circle of radius r_k , all three circles concentric. Setting $r_1 = r_2 = r_3$ must return $3/4$, and it does. The general answer, Theorem 4.1 below, is a sum of three terms, each of which is the probability that a weighted sum of two independent arcsine random variables clears a threshold. The formula is a single one-dimensional integral to evaluate, it reduces to closed forms in several subfamilies, and it makes the qualitative structure visible: which vertex can carry the obtuse angle, how low the obtuse probability can go, and how it behaves as the radii spread apart.

Random triangles have a large literature, including shape-space and random matrix treatments (Edelman and Strang 2015). The concentric-circle family is a two-parameter deformation of the classical circular case that stays elementary throughout: the tools are the law of sines and a change of variables.

2 Setup

Write O for the common center of the three circles and let $r_k \geq 0$ denote the radius of the circle carrying vertex k , with at most one of the three radii equal to zero. (Two zero radii would place two vertices at O and there would be no triangle.) Let $\Theta_1, \Theta_2, \Theta_3$ be independent and uniform on $[0, 2\pi)$ and set

$$P_k = r_k (\cos \Theta_k, \sin \Theta_k), \quad k = 1, 2, 3,$$

so P_k is uniform on the circle of radius r_k about O . Throughout, $\{i, j, k\}$ denotes $\{1, 2, 3\}$ in some order. Scaling all three radii by a common factor scales the triangle without changing any of its angles, so only the ratios $r_1 : r_2 : r_3$ matter.

Let p_k be the probability that the interior angle at P_k is obtuse, and let $P(\text{obtuse}) = \Pr(\text{the triangle } P_1P_2P_3 \text{ is obtuse})$.

Lemma 2.1 (Additivity). $P(\text{obtuse}) = p_1 + p_2 + p_3$.

Proof. The three interior angles sum to π , so at most one of them exceeds $\pi/2$ and the three events are disjoint. Their union is the event that the triangle is obtuse, up to the event that some angle equals exactly $\pi/2$. The angle at P_k is right when

$$D_k = (P_i - P_k) \cdot (P_j - P_k) = r_i r_j \cos(\Theta_i - \Theta_j) - r_i r_k \cos(\Theta_i - \Theta_k) - r_j r_k \cos(\Theta_j - \Theta_k) + r_k^2$$

vanishes. As a function on the torus D_k is real analytic, and it is not identically zero. Its mean over the torus is r_k^2 , since each of the three cosine terms integrates to zero, and that is positive unless $r_k = 0$; when $r_k = 0$ it reduces to $r_i r_j \cos(\Theta_i - \Theta_j)$, which is not identically zero because at most one radius vanishes. The zero set of a nonvanishing real analytic function has measure zero. The same argument disposes of the collinear and coincident configurations, on which “obtuse” is undefined: both are cut out by real analytic equations that do not vanish identically. $\square$

Lemma 2.1 is what makes the problem tractable. It converts a question about a triangle into three separate questions about a single vertex, each of which involves only two independent angles.

3 The angle at one vertex

Fix the vertex P_k and one companion P_i . The angle at P_k is obtuse exactly when $(P_i - P_k) \cdot (P_j - P_k) < 0$, that is, when P_j falls in the open half plane H bounded by the line through P_k perpendicular to P_kP_i , on the side away from P_i (Figure 3). Since P_j is uniform on its own circle, the conditional probability is just the fraction of that circle lying in H , and that fraction depends on P_k and P_i only through the signed distance from the center to the boundary line.

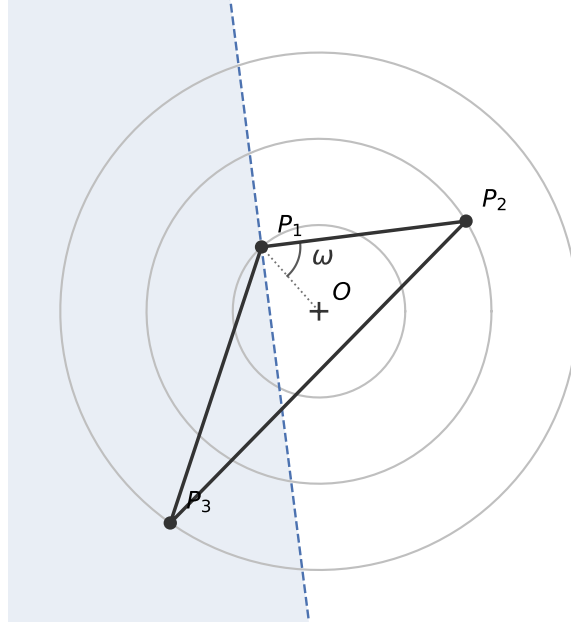


Figure 3: One realization with radii $(1, 2, 3)$. The angle at P_1 is obtuse exactly when P_3 falls in the shaded half plane, whose boundary passes through P_1 perpendicular to P_1P_2 .

Let

$$\omega = \angle OP_k P_i \in [0, \pi]$$

be the angle at P_k in the triangle OP_kP_i ; it is the angle subtended at P_k between the center and the companion vertex. The unit normal of H pointing away from P_i is $n = -(P_i - P_k)/\|P_i - P_k\|$, and because ω is measured between the

directions $P_k \rightarrow O$ and $P_k \rightarrow P_i$, we have $P_k \cdot n = r_k \cos \omega$. So H contains the arc of the circle of radius r_j on which $r_j \cos \Xi < -r_k \cos \omega$ for Ξ uniform, giving

$$\Pr(\text{angle at } P_k \text{ is obtuse} \mid P_i) = \frac{1}{\pi} \arccos \left(\left\lceil \frac{r_k \cos \omega}{r_j} \right\rceil \right) = \frac{1}{2} - \frac{1}{\pi} \arcsin \left(\left\lceil \frac{r_k \cos \omega}{r_j} \right\rceil \right), \quad (1)$$

where $[x] = \max(-1, \min(1, x))$ truncates to the domain of $\arcsin$; the truncation is what happens when the circle of radius r_j lies entirely on one side of the line.

What remains is the distribution of ω .

Lemma 3.1 (The viewing angle). *Let $\psi = \angle OP_i P_k$ and let $h = r_k \sin \omega = r_i \sin \psi$ be the distance from the center to the line $P_k P_i$. Then*

- if $r_k \leq r_i$, the folded angle $\min(\omega, \pi - \omega)$ is uniform on $(0, \pi/2)$;
- if $r_k \geq r_i$, the acute branch $\psi_0 = \min(\psi, \pi - \psi)$ is uniform on $(0, \pi/2)$. (The line meets the circle twice, and ψ is obtuse at the far point, so ψ itself is not confined to $(0, \pi/2)$.)

In both cases $h = \min(r_i, r_k) \sin \mu$ for a μ uniform on $(0, \pi/2)$.

Proof. Let $A = \angle P_k O P_i$ be the central angle. Since $\Theta_i - \Theta_k$ is uniform on the circle, A is uniform on $(0, \pi)$. The interior angles satisfy $A + \omega + \psi = \pi$, and the law of sines gives $r_i \sin \psi = r_k \sin \omega$, so $\sin \psi = (r_k/r_i) \sin \omega$.

If $r_k \leq r_i$ then ψ is opposite the shorter side and is therefore acute, so $\psi = \arcsin((r_k/r_i) \sin \omega)$ is a function of ω alone and $\omega = \pi - A - \psi(\omega)$. Hence

$$\Pr(\omega \leq w) = \Pr(A \geq \pi - w - \psi(w)) = \frac{w + \psi(w)}{\pi}, \quad 0 \leq w \leq \pi. \quad (2)$$

Both $\omega \leq w$ and $\omega \geq \pi - w$ contribute to $\min(\omega, \pi - \omega) \leq w$ for $w \leq \pi/2$, and adding the two pieces from Equation 2 gives $2w/\pi$, because $\psi(w) = \psi(\pi - w)$.

If $r_k \geq r_i$ the line from P_k meets the circle of radius r_i twice, at the near point (where ψ is acute) and the far point (where ψ is obtuse), and $\omega \leq \arcsin(r_i/r_k) \leq \pi/2$. Writing ψ_0 for the acute value, the two branches contribute $\Pr(A \geq \pi - \omega - \psi_0)$ and $\Pr(A \leq \psi_0 - \omega)$, which sum to $2\psi_0/\pi$. $\square$

Lemma 3.1 has a consequence beyond its use here: the distance from the center to the chord through a fixed point and a uniform point on a circle has the law $\min(r_i, r_k) \sin \mu$ with μ uniform, and so depends on the two radii only through their minimum. I record that in passing and claim no priority for it; random chords are old ground in integral geometry and I have not searched that literature. Evaluating Equation 1 needs slightly more than h , however, since it needs the sign of $\cos \omega$ as well. That is the content of the next lemma, which does the bookkeeping once so the theorem can be read off.

Lemma 3.2 (Signed transfer). *Let $S = \text{sign}(\cos \omega)$. For every bounded measurable ϕ ,*

$$\mathbb{E}[S \phi(h)] = \frac{2}{\pi} \int_0^{\pi/2} \phi(r_i \sin t) \mathbf{1}\{r_i \sin t < r_k\} dt. \quad (3)$$

Proof. Suppose first $r_k \geq r_i$. Then $\omega \leq \pi/2$, so $S \equiv 1$, and $h = r_i \sin \psi \leq r_i \leq r_k$ with ψ uniform on $(0, \pi/2)$ by Lemma 3.1; the indicator holds almost surely and Equation 3 is immediate.

Suppose instead $r_k < r_i$ and write G for the distribution function Equation 2, so that $\pi dG = d\omega + d\psi$. Split the integral accordingly. The $d\omega$ part is

$$\frac{1}{\pi} \int_0^\pi \text{sign}(\cos \omega) \phi(r_k \sin \omega) d\omega = 0,$$

because that component has constant density $1/\pi$ and so is preserved by $\omega \mapsto \pi - \omega$, an involution that reverses the sign of $\cos \omega$ while fixing $\sin \omega$. In the $d\psi$ part, ψ increases from 0 to $\arcsin(r_k/r_i)$ as ω runs over $(0, \pi/2)$, where $S = +1$, and decreases back to 0 on $(\pi/2, \pi)$, where $S = -1$; reversing the orientation of the second piece shows that the two contribute equally. Since $r_k \sin \omega = r_i \sin \psi$,

$$\mathbb{E}[S \phi(h)] = \frac{2}{\pi} \int_0^{\arcsin(r_k/r_i)} \phi(r_i \sin t) dt,$$

and $r_i \sin t < r_k$ is exactly $t < \arcsin(r_k/r_i)$. $\square$

4 The main result

Theorem 4.1 (Obtuse probability on concentric circles). *Let X and Y be independent and uniform on $(0, \pi/2)$. Then*

$$p_k = \frac{1}{2} \Pr(r_i^2 \sin^2 X + r_j^2 \sin^2 Y > r_k^2), \quad (4)$$

and consequently

$$P(\text{obtuse}) = \frac{1}{2} \sum_{k=1}^3 \Pr(r_i^2 \sin^2 X + r_j^2 \sin^2 Y > r_k^2). \quad (5)$$

Proof. Take expectations in Equation 1. Since $r_k \cos \omega = S\sqrt{r_k^2 - h^2}$ and arcsin is odd, the truncated arcsine factors as $S\phi(h)$ with

$$\phi(h) = \arcsin \left[\min \left(1, \frac{\sqrt{\max(r_k^2 - h^2, 0)}}{r_j} \right) \right].$$

Lemma 3.2 then gives

$$p_k = \frac{1}{2} - \frac{2}{\pi^2} \int_0^{\pi/2} \arcsin \left[\min \left(1, \frac{\sqrt{r_k^2 - r_i^2 \sin^2 t}}{r_j} \right) \right] \mathbf{1}_{\{r_i \sin t < r_k\}} dt. \quad (6)$$

For fixed t with $r_i \sin t < r_k$,

$$\frac{2}{\pi} \arcsin \left[\min \left(1, \frac{\sqrt{r_k^2 - r_i^2 \sin^2 t}}{r_j} \right) \right] = \Pr(r_j \sin Y < \sqrt{r_k^2 - r_i^2 \sin^2 t}) = \Pr(r_i^2 \sin^2 t + r_j^2 \sin^2 Y < r_k^2),$$

and when $r_i \sin t \geq r_k$ that probability is zero, which absorbs the indicator. Integrating over t against the uniform law of X turns Equation 6 into $p_k = \frac{1}{2} - \frac{1}{2} \Pr(r_i^2 \sin^2 X + r_j^2 \sin^2 Y < r_k^2)$, which is Equation 4. Lemma 2.1 gives Equation 5. $\square$

First, Equation 4 is symmetric in i and j , as it must be, even though the derivation broke that symmetry by conditioning on P_i .

Second, X and Y are auxiliary. They are not the two viewing angles of the same realized triangle, and the corresponding event is not a pointwise restatement of “the angle at P_k is obtuse.” It is an identity between probabilities, not between events. Equation 4 is reached by integrating the companion vertex out through Lemma 3.2 rather than by any pathwise coupling, so there is no realization in which X and Y are the two viewing angles.

Third, if $U = \sin^2 X$ then U has the arcsine distribution on $[0, 1]$, with density $1/(\pi\sqrt{u(1-u)})$, so Equation 5 can be written with U and V independent and identically distributed (iid),

$$P(\text{obtuse}) = \frac{1}{2} \sum_{k=1}^3 \Pr(r_i^2 U + r_j^2 V > r_k^2), \quad U, V \stackrel{\text{iid}}{\sim} \text{Arcsine}(0, 1). \quad (7)$$

The arcsine law is the standard companion of uniform angles (M. G. Kendall and Moran 1963), and in this form the three terms are visibly the same function of the squared radii evaluated at three different thresholds.

For computation, Equation 6 is a one-dimensional integral of an elementary function with two kinks, at $t = \arcsin \min(1, r_k/r_i)$ and where the arcsine saturates. Adaptive quadrature with those break points supplied is accurate to a few units in the last place: it reproduces the two closed forms of Section 5 to within 4×10^{-16} , and its own reported error estimate stays below 3×10^{-9} over a grid of radii.

5 Special cases

Corollary 5.1 (Equal radii). *If $r_1 = r_2 = r_3$ then $p_k = 1/4$ for each k and $P(\text{obtuse}) = 3/4$.*

Proof. With a common radius the event in Equation 4 is $\sin^2 X + \sin^2 Y > 1$, that is $\cos^2 X < \sin^2 Y$, that is $X + Y > \pi/2$, which has probability $1/2$ by symmetry. $\square$

This is the classical answer, recovered without appealing to the inscribed-angle argument. The two routes agree: for a common circle, acute is the same as containing the center, and three uniform points contain the center with probability $1/4$ (Wendel 1962). That second route does not survive the generalization, and it repays being precise about which half of it fails.

Proposition 5.1 (The center and the circumcenter part company). *If $r_1, r_2, r_3 > 0$ then $\Pr(O \in \text{conv}\{P_1, P_2, P_3\}) = 1/4$, whatever the radii. Moreover, with the positivity hypothesis dropped, $\Pr(\text{acute}) = 1 - P(\text{obtuse})$ takes every value in $(0, 1/2]$ as the radii range over all admissible triples.*

Proof. Because $r_k > 0$, the point P_k and the unit vector $u_k = P_k/r_k$ lie on the same open ray from O , hence strictly on the same side of every line through O . So O lies in the convex hull of P_1, P_2, P_3 if and only if u_1, u_2, u_3 fail to lie in a common open half plane, an event determined by $\Theta_1, \Theta_2, \Theta_3$ alone. Those angles are uniform whatever the radii, and Wendel's theorem (Wendel 1962) gives $2^{-2} \left[\binom{2}{0} + \binom{2}{1} \right] = 3/4$ for the probability that three uniform directions in the plane do lie in a half plane, leaving $1/4$ for the containment. The second claim carries no positivity hypothesis, and it needs none: Proposition 5.2 below confines $\Pr(\text{acute})$ to $(0, 1/2]$, and along the slice $(1, 1, \rho)$ the obtuse probability is a continuous function of ρ that equals $1/2$ at $\rho = 0$ and tends to 1 as $\rho \rightarrow \infty$, so every intermediate value is attained. $\square$

The endpoint $\rho = 0$ of that slice is precisely the configuration the first claim excludes. With a vertex sitting at O the containment probability jumps from $1/4$ to 1 , so the two halves of the proposition are describing genuinely different regimes rather than the same one twice.

A triangle is acute exactly when it contains its circumcenter, and the circumcenter is the common center of the circles only when the three radii agree. Unequal radii pull those two points apart, and the two probabilities follow suit: containment is pinned at $1/4$ by the directions alone, while acuteness responds to the radii and ranges over the whole of $(0, 1/2]$. Equation 5 is the measurement of how far the two have separated. They agree numerically on the level set $P(\text{obtuse}) = 3/4$, which is a curve rather than a point, but the underlying events coincide only at equal radii.

Corollary 5.2 (Which vertex can be obtuse). *$p_k > 0$ if and only if $r_k^2 < r_i^2 + r_j^2$, and $p_k = 1/2$ if and only if $r_k = 0$. At most one of the three inequalities can fail, except when one radius is zero and the other two are equal, in which case two fail.*

Proof. The random variable $r_i^2 \sin^2 X + r_j^2 \sin^2 Y$ has support $[0, r_i^2 + r_j^2]$ with positive density in the interior, so the probability in Equation 4 is positive exactly when $r_k^2 < r_i^2 + r_j^2$ and equals one exactly when $r_k = 0$. If two of the inequalities failed, say $r_k^2 \geq r_i^2 + r_j^2$ and $r_j^2 \geq r_i^2 + r_k^2$, adding them forces $r_i = 0$ and then $r_j = r_k$. $\square$

So the vertex sitting on a circle that dominates the other two in the Pythagorean sense is never the obtuse one, and the sum in Equation 5 has only two nonzero terms whenever the radii are spread out. Being the largest is not enough on its own, though: at radii $(1, 1, 1.2)$ the outer vertex still carries the obtuse angle with probability 0.1058.

Corollary 5.3 (One vertex at the center). *If $r_3 = 0$ and $r_1 \leq r_2$ then*

$$P(\text{obtuse}) = 1 - \frac{1}{\pi} \arcsin\left(\frac{r_1}{r_2}\right).$$

Proof. By Corollary 5.2, $p_3 = 1/2$, and $p_2 = 0$ because $r_2^2 \geq r_1^2 + 0$. For the remaining term, $p_1 = \frac{1}{2} \Pr(r_2^2 \sin^2 X > r_1^2) = \frac{1}{2} \left(1 - \frac{2}{\pi} \arcsin(r_1/r_2)\right)$. $\square$

The expression decreases from 1 at $r_1/r_2 \rightarrow 0$ to $1/2$ at $r_1 = r_2$. The lower end is the configuration of Proposition 5.2 below.

Corollary 5.4 (Two equal radii). *If $r_1 = r_2 = 1$ and $r_3 = \rho \geq \sqrt{2}$ then*

$$P(\text{obtuse}) = 1 - \frac{4}{\pi^2} \chi_2\left(\frac{1}{\rho}\right), \quad \chi_2(s) = \sum_{n \geq 0} \frac{s^{2n+1}}{(2n+1)^2},$$

where χ_2 is the Legendre chi function.

Proof. Corollary 5.2 gives $p_3 = 0$. For p_1 (and by symmetry p_2), Equation 4 asks for $\Pr(\sin^2 X + \rho^2 \sin^2 Y > 1) = \Pr(\rho \sin Y > \cos X)$, and conditioning on X gives

$$p_1 = \frac{1}{2} \left(1 - \frac{4}{\pi^2} \int_0^{\pi/2} \arcsin\left(\frac{\cos x}{\rho}\right) dx\right).$$

Expanding $\arcsin$ in its Taylor series and integrating term by term with Wallis' formula, $\int_0^{\pi/2} \cos^{2n+1} x dx = (2n)!!/(2n+1)!!$, collapses the coefficients to $\int_0^{\pi/2} \arcsin(s \cos x) dx = \chi_2(s)$. $\square$

The per-vertex expression for p_1 holds on the wider range $\rho \geq 1$, which is where $\arcsin(\cos x/\rho)$ stays clear of its truncation; the hypothesis $\rho \geq \sqrt{2}$ additionally kills the third term, and is therefore the single condition under which the displayed total is valid. On the wider range $\chi_2(1) = \pi^2/8$ returns $p_1 = 1/4$ at $\rho = 1$, consistent with Corollary 5.1. Parity is what puts χ_2 here rather than the dilogarithm itself: $\chi_2(s) = \frac{1}{2} [\text{Li}_2(s) - \text{Li}_2(-s)]$ is the odd part of Li_2 , and $\arcsin$ being an odd function leaves only odd powers of s to integrate.

Proposition 5.2 (Bounds). $1/2 \leq P(\text{obtuse}) < 1$. The lower bound is attained exactly when one radius is zero and the other two are equal, and $P(\text{obtuse}) \rightarrow 1$ as the radii separate.

Proof. Write $a \leq b \leq c$ for the sorted values of r_1^2, r_2^2, r_3^2 , let U, V be independent arcsine variables as in Equation 7, and let $q_a = \Pr(bU + cV > a)$, $q_b = \Pr(aU + cV > b)$, and $q_c = \Pr(aU + bV > c)$ be the corresponding terms of Equation 7, so that $2P(\text{obtuse}) = q_a + q_b + q_c$.

The arcsine law is symmetric about $1/2$, so (U, V) and $(1 - U, 1 - V)$ have the same distribution and

$$q_b = \Pr(a(1 - U) + c(1 - V) > b) = \Pr(aU + cV < a + c - b).$$

Let $E_1 = \{bU + cV \leq a\}$ and $E_2 = \{aU + cV < a + c - b\}$, so that $q_a + q_b = 1 - \Pr(E_1) + \Pr(E_2)$. On E_1 we have $aU + cV \leq bU + cV \leq a \leq a + c - b$, with at least one inequality strict unless $a = b = c$; hence $E_1 \subseteq E_2$ up to a null set and $q_a + q_b \geq 1$. Dropping $q_c \geq 0$ gives $P(\text{obtuse}) \geq 1/2$. (When $a = b = c$ the three terms are each $1/2$ and the bound is not tight.)

Equality forces $q_c = 0$ and $\Pr(E_2 \setminus E_1) = 0$. Suppose $a > 0$. If $c > b$, take U close to 1 and V close to 0 subject to $cV > a(1 - U)$. Then $bU + cV > bU + a(1 - U) \geq a$, so the point is outside E_1 , while $aU + cV$ is close to a , which is strictly below $a + c - b$, so it lies in E_2 . If instead $c = b > a$, reverse the constraint to $cV < a(1 - U)$. Then $aU + cV < aU + a(1 - U) = a = a + c - b$ puts the point in E_2 , while $bU + cV$ is close to $c > a$ and so is outside E_1 . Both prescriptions describe sets of positive Lebesgue measure, on which the arcsine density is positive, so $\Pr(E_2 \setminus E_1) > 0$ in either case.

That leaves $a = b = c$ as the only configuration with $a > 0$ not yet excluded, and there $q_c = 1/2$, so equality fails through the dropped term instead. Hence $a = 0$.

It is tempting to finish by saying that $q_c = 0$ then forces $b = c$, but $q_c = \Pr(bV > c)$ vanishes for every $b \leq c$ once $a = 0$, so it carries no information. The other equality condition does the work. At $a = 0$ the set $E_1 = \{bU + cV \leq 0\}$ is null, while $E_2 = \{cV < c - b\}$ has positive probability whenever $b < c$; so $\Pr(E_2 \setminus E_1) = 0$ forces $b \geq c$, and with $b \leq c$ that gives $b = c$. At $a = 0$ with $b = c$ we have $q_a = 1$, $q_b = 0$, and $q_c = 0$.

For the upper bound, fix the radii and work in the angle torus, where the law is uniform. The three dot products are continuous in $(\Theta_1, \Theta_2, \Theta_3)$, so the acute configurations form an open subset of the torus, and exhibiting a single one gives it positive probability. Order the radii so that $r_3 = \max$.

If all three are positive and not all equal, place

$$P_1 = (0, r_1), \quad P_2 = (0, -r_2), \quad P_3 = (r_3, 0).$$

The three vertex dot products are $r_1(r_1 + r_2)$, $r_2(r_1 + r_2)$, and $r_3^2 - r_1r_2$. The first two are positive, and the third is positive because $r_3 = \max$ gives $r_3^2 \geq r_1r_2$ with equality only when $r_1 = r_2 = r_3$. If instead all three are equal, three points $2\pi/3$ apart give dot products $3r^2/2 > 0$. If one radius vanishes, say $r_1 = 0 < r_2 \leq r_3$, put the other two at angle θ with $0 < \cos \theta < r_2/r_3$; the dot products are then $r_2r_3 \cos \theta$, $r_2(r_2 - r_3 \cos \theta)$, and $r_3(r_3 - r_2 \cos \theta)$, all positive. So $P(\text{obtuse}) < 1$ in every case, and Corollary 5.3 shows values arbitrarily close to 1, as $r_1/r_2 \rightarrow 0$ with $r_3 = 0$. $\square$

Name the extremal configuration: one vertex pinned at the center and the other two on a common circle. There the triangle is isosceles, both base angles are acute by construction, and the triangle is obtuse exactly when the central angle exceeds $\pi/2$, which happens half the time. No arrangement of concentric circles does better than an even split.

At the other end of the range sits the equal-radius configuration, which is a local maximum but not a smooth one.

Proposition 5.3 (Local maximum at equal radii). Normalize the squared radii to $a_k = r_k^2$ with $a_1 + a_2 + a_3 = 3$ and write $a_k = 1 + \alpha_k$. Then, as $\alpha \rightarrow 0$,

$$P(\text{obtuse}) = \frac{3}{4} - \frac{1}{4\pi^2} \sum_{k < l} (\alpha_k - \alpha_l)^2 \log \frac{1}{|\alpha_k - \alpha_l|} + O(|\alpha|^2), \quad (8)$$

where a vanishing difference contributes nothing to the sum. In particular $3/4$ is a strict local maximum over radius ratios, attained only at $r_1 = r_2 = r_3$, and P is differentiable but not twice differentiable there. The remainder is uniform in the direction of α ; Lemma A.1 supplies the exact endpoint decomposition on which that rests.

Proof. Condition on X in Equation 4 and use $\Pr(\sin^2 Y \leq z) = \frac{2}{\pi} \arcsin \sqrt{z}$ to write $q_k = 2p_k$ as a single integral,

$$q_k = 1 - \frac{4}{\pi^2} I_k, \quad I_k = \int_0^{\pi/2} \arccos \left(\min \left(1, \sqrt{[\mu \sin^2 x - \eta]^+} \right) \right) dx, \quad (9)$$

with $\mu = a_i/a_j$ and $\eta = (a_k - a_j)/a_j$, and $[\cdot]^+$ the positive part. At $\alpha = 0$ the integrand is $\arccos(\sin x) = \pi/2 - x$ and $I_k = \pi^2/8$.

The integrand of Equation 9 is a smooth function of (α, x) on any interval $[\delta, \pi/2 - \delta]$, so only the two endpoints can produce behavior worse than $O(|\alpha|)$. Near $x = 0$ both arguments of $\arccos$ are small, $\arccos t - \arccos t_0 = t_0 - t + O(t^3)$, and $\sin x = x + O(x^3)$, so that endpoint contributes

$$\int_0^\delta \left(x - \sqrt{[\mu x^2 - \eta]^+} \right) dx = \frac{\eta}{4\sqrt{\mu}} \log \frac{1}{|\eta|} + O(|\alpha|) = \frac{\eta}{4} \left(1 - \frac{\lambda}{2} \right) \log \frac{1}{|\eta|} + O(|\alpha|),$$

where $\lambda = \mu - 1 = (a_i - a_j)/a_j$; the logarithm appears because the integrand is cut off at $x \asymp \sqrt{|\eta|}$ rather than at $|\eta|$. Substituting $x \mapsto \pi/2 - x$ gives the same expression at the other endpoint with i and j interchanged.

Now sum over k and over the two endpoints, which is a sum over ordered pairs. Writing $e_{kl} = \alpha_k - \alpha_l$, we have $\eta = e_{kj}(1 + O(|\alpha|))$ and $\lambda = e_{ij}(1 + O(|\alpha|))$, so the ordered pair (k, j) contributes $\frac{1}{4} \log(1/|e_{kj}|) [e_{kj} - e_{kj}\alpha_j - \frac{1}{2}e_{kj}e_{ij}]$ up to $O(|\alpha|^2)$. Pair (k, j) with (j, k) , which share the logarithm and the third index i . Because $e_{jk} = -e_{kj}$, the terms linear in e cancel; the second terms combine to $-e_{kj}\alpha_j - e_{jk}\alpha_k = e_{kj}(\alpha_k - \alpha_j) = e_{kj}^2$; and the third terms combine to $-\frac{1}{2}(e_{kj}e_{ij} + e_{jk}e_{ik}) = \frac{1}{2}e_{kj}(e_{ik} - e_{ij}) = -\frac{1}{2}e_{kj}^2$. Hence

$$\sum_{k=1}^3 I_k = \frac{3\pi^2}{8} + \frac{1}{8} \sum_{k < l} e_{kl}^2 \log \frac{1}{|e_{kl}|} + \Lambda(\alpha) + O(|\alpha|^2),$$

where Λ collects the smooth contributions and is differentiable at $\alpha = 0$. Since P is symmetric in the three radii, $\nabla \Lambda(0)$ is parallel to $(1, 1, 1)$, and P is constant in that direction by scale invariance, so Λ has no linear term on the normalized slice $\sum_k \alpha_k = 0$. Finally $2P(\text{obtuse}) = \sum_k q_k = 3 - \frac{4}{\pi^2} \sum_k I_k$, which is Equation 8.

Because $t^2 \log(1/t)$ dominates t^2 as $t \downarrow 0$, the displayed sum controls the remainder, and it is strictly positive unless all three α_k agree. The same domination shows the second directional derivative diverges in every direction other than the dilation. $\square$

Two steps of that proof were quoted rather than carried out: the uniformity of the remainder in the direction of α , and the differentiability of the smooth part Λ . Both follow from the endpoint integral being elementary, so that its singular part separates exactly instead of asymptotically. Section A does the separation, and settles a cubic correction that is not as negligible as it looks.

Figure 4 shows the equal-radius maximum along one slice. It is a local maximum only: pushing the radii apart drives the probability toward 1, and $(1, 5, 25)$ already gives 0.9353.

The value $3/4$ does not characterize equal radii. The level set $P(\text{obtuse}) = 3/4$ is a curve in the two-dimensional space of radius ratios that passes through the equal-radius point; for instance the triple $(1, 2, r_3)$ reaches $3/4$ at $r_3 = 2.9188$.

The obtuse probability is also not monotone in the radii. Figure 4 shows the slice $(1, 1, \rho)$: it rises to $3/4$ at $\rho = 1$, falls to a local minimum at $\rho = \sqrt{2}$, and rises again toward 1. The corner at $\sqrt{2}$ is exactly where the third term of Equation 5 switches off, by Corollary 5.2, and the kink is a consequence of that term hitting the boundary of its support rather than anything smooth going on in the other two. The corner has an exact size, and it is a jump in the derivative rather than in the value: just below the threshold the arcsine density near the top of its support gives $p_3 \approx (2 - \rho^2)/(2\pi)$, so the right derivative exceeds the left by $\sqrt{2}/\pi$. Numerically the two one-sided slopes are -0.19758 and 0.25258 , a jump of 0.45016 against $\sqrt{2}/\pi = 0.45016$. Figure 5 shows the full two-parameter picture, with the same ridge running through it.

Table 1 collects the section. The first three rows are the subfamilies where Equation 5 closes, and the last two are the range it takes over everything else; setting $r_1 = r_2$ in the second row returns $1/2$, which is the minimum in the fourth.

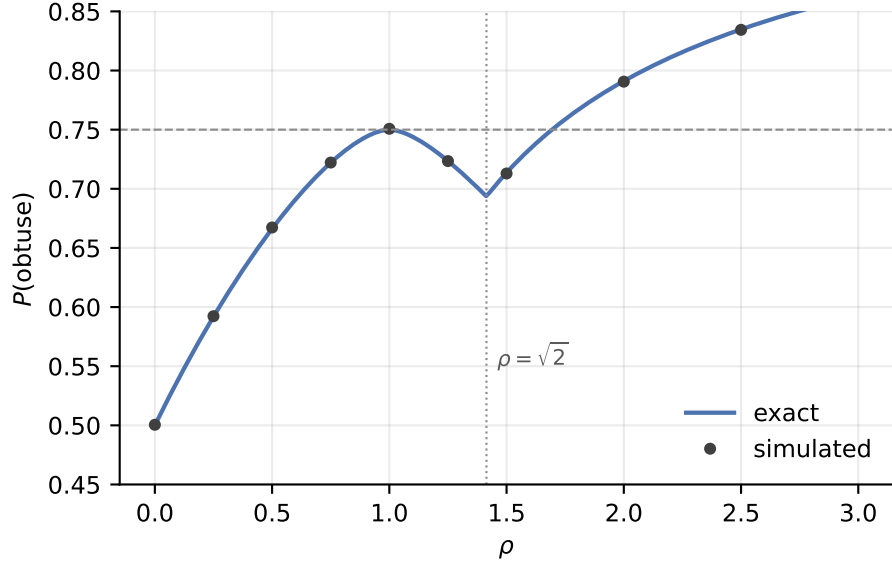


Figure 4: Obtuse probability along the slice $(1, 1, \rho)$. The curve is Equation 5; the points are Monte Carlo estimates from 10^6 triangles each. The dashed lines mark the equal-radius value $3/4$ at $\rho = 1$ and the threshold $\rho = \sqrt{2}$ beyond which the third vertex can no longer be the obtuse one.

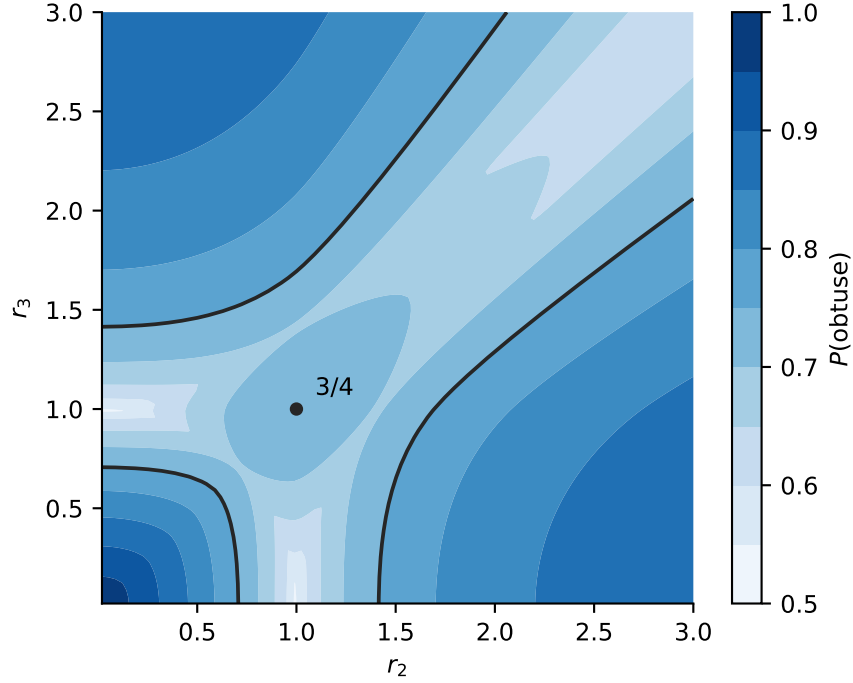


Figure 5: Obtuse probability over the two-parameter family with $r_1 = 1$. The heavy contour is the level set $3/4$, which passes through the equal-radius point (marked) but is not confined to it. The probability falls to $1/2$ only in the corner where one radius vanishes and the other two agree, and rises toward 1 as the radii separate.

Table 1: Closed forms and bounds for the special cases of Equation 5.

Configuration	$P(\text{obtuse})$	
$r_1 = r_2 = r_3$	$3/4$	Corollary 5.1
$r_3 = 0, r_1 \leq r_2$	$1 - \frac{1}{\pi} \arcsin \frac{r_1}{r_2}$	Corollary 5.3
$r_1 = r_2 = 1, r_3 = \rho \geq \sqrt{2}$	$1 - \frac{4}{\pi^2} \chi_2\left(\frac{1}{\rho}\right)$	Corollary 5.4
any admissible radii	$1/2 \leq P < 1$	Proposition 5.2
$r_3 = 0, r_1 = r_2$	$1/2$, the minimum	Proposition 5.2

6 Every rotationally symmetric scheme is a mixture

Fixed concentric circles look like one sampling convention among the many that Carroll’s question admits, but they are the conditional kernel for the whole rotationally symmetric class, which turns Equation 5 into a statement about every such convention at once.

Proposition 6.1 (Mixtures over the radii). *Let P_1, P_2, P_3 be independent, each with a distribution invariant under rotation about O , and let $R_k = \|P_k\|$ with $\Pr(R_k = 0) = 0$ for at least two of the three. Writing $P_{\text{obt}}(r_1, r_2, r_3)$ for the fixed-radius probability of Equation 5,*

$$P(\text{obtuse}) = \mathbb{E}[P_{\text{obt}}(R_1, R_2, R_3)]. \quad (10)$$

Consequently the obtuse probabilities achievable across the entire class are exactly the interval $[1/2, 1)$.

Proof. A rotationally invariant law on the plane factors as $P_k = R_k(\cos \Theta_k, \sin \Theta_k)$ with Θ_k uniform on $[0, 2\pi)$ and independent of R_k . Independence across k makes $(\Theta_1, \Theta_2, \Theta_3)$ uniform on the torus and independent of (R_1, R_2, R_3) , so conditionally on the radii the configuration is exactly the one in Section 2. Conditioning and averaging gives Equation 10.

For the range, Proposition 5.2 confines the integrand to $[1/2, 1)$, so $P(\text{obtuse}) \geq 1/2$ for every scheme in the class. The upper end needs a little more than the convex hull, whose closure would readmit 1: since $1 - P_{\text{obt}}(R_1, R_2, R_3)$ is strictly positive almost surely and bounded, its expectation is strictly positive, so $P(\text{obtuse}) < 1$. Both ends are approached by deterministic radii: along the slice $(1, 1, \rho)$ the probability is continuous in ρ , equals $1/2$ at $\rho = 0$ by Corollary 5.3, and tends to 1 as $\rho \rightarrow \infty$ by Corollary 5.4, so the intermediate value theorem supplies everything between. $\square$

The Gaussian convention is the case to test this against, and Equation 10 does better than reproduce its value: it evaluates in closed form.

Corollary 6.1 (The Gaussian convention). *Let R_1, R_2, R_3 be independent Rayleigh variables, so that the three vertices are independent standard bivariate normals. Then $\mathbb{E}[P_{\text{obt}}(R_1, R_2, R_3)] = 3/4$.*

Proof. For a standard bivariate normal $R_k^2/2$ is $\text{Exp}(1)$, and only ratios matter, so set $E_k = R_k^2/2$ and replace R_k^2 by E_k throughout. By Equation 7 and exchangeability of the E_k ,

$$P(\text{obtuse}) = \frac{1}{2} \sum_k \Pr(E_i U + E_j V > E_k) = \frac{3}{2} \Pr(E_1 U + E_2 V > E_3).$$

Condition on U, V, E_1, E_2 . Since $\Pr(E_3 > s) = e^{-s}$ and $\mathbb{E}[e^{-uE}] = 1/(1+u)$ for $E \sim \text{Exp}(1)$,

$$\Pr(E_1 U + E_2 V > E_3) = 1 - \mathbb{E}[e^{-E_1 U}] \mathbb{E}[e^{-E_2 V}] = 1 - \mathbb{E}\left[\frac{1}{1+U}\right]^2.$$

For U arcsine on $[0, 1]$ the standard integral

$$\int_0^1 \frac{du}{\pi \sqrt{u(1-u)}(u+a)} = \frac{1}{\sqrt{a(a+1)}}$$

gives $\mathbb{E}[1/(1+U)] = 1/\sqrt{2}$ at $a = 1$. The inner probability is therefore $1 - 1/2 = 1/2$, and $P(\text{obtuse}) = 3/4$. $\square$

This supplies a link the literature reports as missing. Edelman and Strang (2015) note in their Section 1.1, and again in Section 3.6, that the uniform-angle convention and the Gaussian convention both give $3/4$, and that they are “not aware

of an argument that links the ‘angle picture’ with the normal distribution”; they point to Portnoy (1994) for related discussion of how often $3/4$ arises. By the identification in Section 1 the angle picture is the equal-radius member of the family that Proposition 6.1 integrates, the Gaussian convention is the integral, and Corollary 6.1 evaluates it. The two agree because $\mathbb{E}[1/(1+U)] = 1/\sqrt{2}$.

Two cautions on how much that says. The equal-radius configuration carries no mass under the Rayleigh law, so it is a member of the family of kernels rather than a component of positive weight. And what the computation matches is the obtuse probability, not the two shape measures, which genuinely differ: the Gaussian law is far from uniform on angle space, as Edelman and Strang (2015) emphasize.

Simulation confirms both sides: 4×10^6 Gaussian triangles give 0.7496 (standard error 0.0002), and averaging Equation 5 over 20,000 Rayleigh radius triples gives 0.7495 (standard error 0.0005). What makes Corollary 6.1 surprising is the spread of what is being averaged. Over those same draws the fixed-radius probability ranged from 0.5141 to 0.9856, so a function that swings across nearly half the unit interval averages back to precisely the value it takes when the three radii coincide.

7 Numerical evidence

7.1 Against the definition

The simulation is a direct transcription of the definition: sample three angles, form the three vertices, and test the sign of the three dot products. No part of the derivation is reused, so agreement is a genuine check.

```
theta = rng.uniform(0, 2 * np.pi, size=(n, 3))
pts = np.stack([radii * np.cos(theta), radii * np.sin(theta)], axis=-1)
np.mean(((pts[:, i] - pts[:, k]) * (pts[:, j] - pts[:, k])).sum(1) < 0)
```

Table 2 compares Equation 5 against $n = 2 \times 10^6$ simulated triangles for a range of radius triples, including the degenerate case with a vertex at the center and cases where one circle dominates. The last column is the standardized difference, $(\hat{p} - p)\sqrt{n/(p(1-p))}$, which should behave like a standard normal draw, and SE is the standard error of the simulated estimate.

Table 2: Exact obtuse probabilities from Equation 5 against Monte Carlo estimates from $n = 2 \times 10^6$ triangles per configuration.

(r_1, r_2, r_3)	Exact	Simulated	SE	z
1, 1, 1	0.750000	0.749823	0.000306	-0.58
1, 1, 0.5	0.666280	0.666729	0.000333	+1.35
1, 1, 1.2	0.731314	0.731217	0.000313	-0.31
1, 1, 2	0.791146	0.791095	0.000287	-0.17
1, 2, 3	0.757947	0.757672	0.000303	-0.90
2, 3, 4	0.711715	0.711651	0.000320	-0.20
0.5, 0.8, 1	0.690747	0.690962	0.000327	+0.66
1, 5, 25	0.935280	0.935172	0.000174	-0.62
1, 2, 0	0.833333	0.833414	0.000264	+0.30
0.3, 1, 1.05	0.604618	0.604937	0.000346	+0.92

All ten standardized differences sit within the range expected from ten independent normal draws. The package’s test suite compares the three per-vertex probabilities separately, which is the stronger check: it exercises not only Equation 5 but also the decomposition in Lemma 2.1 and the individual terms of Equation 4.

7.2 The expansion at equal radii

Table 3 checks Equation 8 directly. The ratio of the exact deficit to the predicted one approaches 1 slowly, like $1 + c/\log(1/\varepsilon)$, which is the signature of an $O(|\alpha|^2)$ remainder sitting under an $|\alpha|^2 \log(1/|\alpha|)$ leading term.

Table 3: The expansion Equation 8 against exact values, for three perturbation directions $\alpha = \varepsilon d$ of the squared radii.

d	ε	Exact deficit	Equation 8	Ratio
(0, 0, 1)	0.01	3.039e-05	2.333e-05	1.3028
(0, 0, 1)	0.001	4.202e-07	3.500e-07	1.2009
(0, 0, 1)	0.0001	5.369e-09	4.666e-09	1.1507
(1, -1, 0)	0.01	8.404e-05	6.297e-05	1.3347
(1, -1, 0)	0.001	1.190e-06	9.796e-07	1.2151
(1, -1, 0)	0.0001	1.540e-08	1.330e-08	1.1585
(3, -1, -2)	0.01	4.818e-04	3.318e-04	1.4519
(3, -1, -2)	0.001	7.247e-06	5.768e-06	1.2564
(3, -1, -2)	0.0001	9.693e-08	8.218e-08	1.1795

7.3 The containment invariant

Proposition 5.1 makes a prediction that is easy to check directly and easy to disbelieve: the triangle surrounds the origin with probability $1/4$ no matter how far apart the radii are pulled.

Simulation bears it out. Over $(1, 1, 1)$, $(1, 1, 0.5)$, and $(1, 5, 25)$, whose obtuse probabilities differ by more than 0.26, the origin fell inside the triangle in 0.2493, 0.2499, and 0.2504 of 10^6 draws each.

8 Software

The `rtcc` package evaluates Equation 6 with adaptive quadrature and the kinks supplied as break points, and includes the simulator used above. Radii are unitless, and a radius of zero is allowed.

```
$ uv run rtcc 1 1 1
radii          1, 1, 1
P(obtuse)      0.750000
  at vertex 1  0.250000
  at vertex 2  0.250000
  at vertex 3  0.250000

$ uv run rtcc 1 2 3 --simulate 1000000 --seed 1
radii          1, 2, 3
P(obtuse)      0.757947
  at vertex 1  0.472162
  at vertex 2  0.285784
  at vertex 3  0.000000
simulated      0.758088 (SE 0.000428)
```

The `--json` flag emits the same values as JavaScript Object Notation (JSON) for scripting, and the Python interface exposes `p_obtuse`, `p_vertices`, and `simulate` directly. The accuracy quoted in Section 4 is checked by the tests `test_vertex_at_the_center` and `test_two_equal_radii_closed_form`, which compare the quadrature against Corollary 5.3 and Corollary 5.4; they assert agreement to 10^{-9} , well inside the few units in the last place actually observed. Every number and figure in this paper is computed at build time from the package rather than transcribed. Source and tests are at <https://github.com/bgreenwell/rtcc>.

9 Discussion

Giving each vertex its own circle keeps everything that made the circular Pillow Problem well posed and adds two free parameters. The answer stays elementary: Equation 5 is a sum of three tail probabilities for weighted sums of independent arcsine variables, and the classical $3/4$ is the point where all three thresholds coincide and each tail is exactly one half.

The geometry that drops out of the algebra is the part I find most useful. Proposition 5.1 isolates what the generalization actually breaks: the probability that the triangle surrounds the common center never moves off $1/4$, and it is the identification of that center with the circumcenter, not the counting argument behind it, that equal radii were quietly supplying. Corollary 5.2 says the obtuse angle can only sit at a vertex whose circle does not dominate the other two in the Pythagorean sense, which is why the sum usually has two nonzero terms rather than three. Proposition 5.2 says

no concentric arrangement can make obtuse triangles rarer than even odds, and identifies the single configuration that achieves it.

The agreement between the circular and Gaussian conventions, which Edelman and Strang (2015) record as lacking a satisfying theoretical link, is a consequence of the mixture identity and a single arcsine integral. That is Corollary 6.1, and it is the result I would lead with. Proposition 6.1 is why the fixed-radius formula is worth more than the family it was derived for. Carroll’s question has no answer until a convention is named, and the mixture identity says that naming a rotationally symmetric convention is the same as naming a distribution for the radii. The floor of $1/2$ then applies to all of them at once, and the agreement between the circle and Gaussian conventions becomes a fact about an average rather than a coincidence between two special cases.

Replacing the circles by spheres in $\mathbb{R}^d$ changes the conditional half-space probability in Equation 1 from an arcsine to an incomplete beta. Two parts of that extension should be distinguished. The mixture identity Proposition 6.1 is a routine transcription: its proof uses only that a rotationally invariant law factors into a radius and an independent uniform direction, which holds in every dimension, so the conditional-kernel reading of the fixed-radius family carries over unchanged. What is genuinely open is the fixed-radius formula itself, since Lemma 3.1 and Lemma 3.2 are planar arguments and I do not know what replaces the law of sines on a sphere. The target is known in the Gaussian case, where Edelman and Strang (2015) give $3(1 - I(3/4, d/2, d/2))$ and random triangles become acute with probability tending to one as d grows (Eisenberg and Sullivan 1996), so a d -dimensional analogue of Equation 5 would have something definite to reproduce. Dropping independence between the three vertices is the other direction, and the harder one, since Lemma 2.1 is all that survives.

Use of generative AI

The author is responsible for this work: the problem, the results, the exposition, and any errors in them.

Two large language models were used in its preparation, both under the author’s direction. `claude-opus-5` (Anthropic), through the Claude Code interface, was used to check and revise proofs, to revise the exposition, and to implement the accompanying Python package, simulations, and figures. `gpt-5.6-iso1` (OpenAI), through the Codex interface, was used independently to review proofs and identify errors; several were found and corrected as a result.

Every numerical result was verified by the author against the accompanying software, and every proof was checked by the author before inclusion.

Acknowledgments

I am grateful to Harry Khamis, who introduced me to Buffon’s needle and whose own extensions of it to concentric circles and radial lines (Khamis 1987, 1991) are where the idea for this paper came from.

A The endpoint expansion

This appendix supplies the two steps Proposition 5.3 quoted, and disposes of a cubic correction that enters at the same order as the term Equation 8 keeps.

Lemma A.1 (Exact endpoint decomposition). *Fix $\delta > 0$ and let*

$$J(\mu, \eta) = \int_0^\delta \left(x - \sqrt{[\mu x^2 - \eta]^+} \right) dx.$$

Then for $|\eta| < \mu\delta^2$,

$$J(\mu, \eta) = \frac{\eta}{4\sqrt{\mu}} \log \frac{1}{|\eta|} + A(\mu, \eta), \quad (11)$$

where A is real analytic on a neighborhood of $(1, 0)$.

Proof. The integrand is elementary. Writing $F(x) = \frac{x}{2} \sqrt{\mu x^2 - \eta} - \frac{\eta}{2\sqrt{\mu}} \log(\sqrt{\mu} x + \sqrt{\mu x^2 - \eta})$ for an antiderivative of $\sqrt{\mu x^2 - \eta}$, integration for $\eta > 0$ splits at $x_0 = \sqrt{\eta/\mu}$, where the positive part switches on, and gives $J = \delta^2/2 - F(\delta) + F(x_0)$. Since $\sqrt{\mu x_0^2 - \eta} = 0$ and $\sqrt{\mu} x_0 = \sqrt{\eta}$,

$$F(x_0) = -\frac{\eta}{2\sqrt{\mu}} \log \sqrt{\eta} = \frac{\eta}{4\sqrt{\mu}} \log \frac{1}{\eta},$$

which is the displayed singular term; the case $\eta < 0$ has no switch and the same computation applies at $x = 0$. What remains, $A = \delta^2/2 - F(\delta)$, involves $\sqrt{\mu\delta^2 - \eta}$ and $\log(\sqrt{\mu}\delta + \sqrt{\mu\delta^2 - \eta})$, both real analytic in (μ, η) wherever $\mu\delta^2 > |\eta|$ and $\mu > 0$. $\square$

Equation 11 is what those two steps needed. Because A is analytic, the endpoint contributions are differentiable in α once the explicit $\eta \log(1/|\eta|)$ terms are removed, which is exactly the differentiability of Λ ; and because A 's derivatives are bounded on a fixed neighborhood, the resulting $O(|\alpha|^2)$ bound holds with a constant independent of the direction of α . Two approximations were used to reach J , namely $\arccos t - \arccos t_0 = t_0 - t + O(t^3)$ and $\sin x = x + O(x^3)$; both are Taylor expansions with remainders uniform on $[0, \delta]$. Since the α_k can be centered, $\max_{k < l} |e_{kl}|$ is comparable to $|\alpha|$, so the displayed sum in Equation 8 is of exact order $|\alpha|^2 \log(1/|\alpha|)$.

The cubic Taylor corrections need one further step, because at first sight they are not small enough to ignore. They lead to integrals of $[\mu x^2 - \eta]_+^{3/2}$, whose singular part is $\frac{3\eta^2}{16\sqrt{\mu}} \log \frac{1}{|\eta|}$ by the same antiderivative used in Lemma A.1, contributing $-\frac{1}{32\sqrt{\mu}} \eta^2 \log \frac{1}{|\eta|}$ to the endpoint. That is the order of the terms Equation 8 retains.

A second correction of the same order cancels it, and the two must be taken together. Reaching J also replaced $\sin x$ by x , and $\sin^2 x = x^2 - x^4/3 + O(x^6)$, so the square root carries an extra $-\mu x^4 / (6\sqrt{\mu x^2 - \eta})$. Now $\int x^4 (x^2 - \eta)^{-1/2} dx$ has log coefficient $3\eta^2/8$, giving $+\frac{1}{32\sqrt{\mu}} \eta^2 \log \frac{1}{|\eta|}$, which is the negative of the first. The two same-order corrections annihilate each other at each endpoint separately, so no $\eta^2 \log(1/|\eta|)$ term ever reaches the sum and the coefficient in Equation 8 is the one Lemma A.1 already gives. The higher terms of both series enter at order $|\alpha|^3 \log(1/|\alpha|)$.

Two checks. Numerically, the even-in- η part of a single endpoint is $-\eta^2/2$ with no logarithm: the ratio of that part to η^2 holds at -0.500 across $\eta = 10^{-3}, 10^{-4}, 10^{-5}$, whereas an uncanceled $\eta^2 \log$ term would make it grow like $\log(1/\eta)$. Structurally, at $\mu = 1$ the reflection $x \mapsto \pi/2 - x$ turns $\sin^2 x$ into $1 - \sin^2 x$, whence $\arccos \sqrt{\cos^2 x - \eta} = \arcsin \sqrt{\sin^2 x + \eta}$, and since $\arcsin u + \arccos u = \pi/2$ pointwise, including where the arguments are clipped,

$$I(1, \eta) + I(1, -\eta) = \frac{\pi^2}{4},$$

so $I(1, \cdot) - \pi^2/8$ is odd and every even-order term in η , the $\eta^2 \log$ term included, vanishes identically there.

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